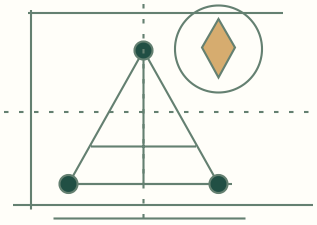
Independent academic prototype



Whitney Learning Compass

From Curriculum Goals to Hands-On Discovery

An evidence-grounded decision assistant that connects a teacher's learning goals to real museum workshops and an editable learning journey.

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MGT 409 · AI Foundations for Managers · Fall 2026
October 23, 2026

GitHub repository
https://github.com/ql200043-art/whitney-learning-compass

App / demonstration
Local React + FastAPI prototype. No public deployment. Historical live workflow documented on pages 5–6.

Whitney Learning Compass is an independent academic prototype developed using publicly available Eli Whitney Museum & Workshop program information. It is not an official museum product, booking platform, or museum-endorsed service.

The question behind the project
Can a bounded AI workflow make museum-program selection more informed and actionable while preserving teacher judgment and visible uncertainty?

Source foundation: official School Programs catalog.

A personal connection became a practical planning question.

My fellowship match prompted me to explore how hands-on museum experiences might connect to classroom goals.

I am a Yale SOM MBA/MEM joint-degree student, and before Yale I worked in sustainability consulting at PwC. That background has made evidence, organizational decisions and practical implementation familiar parts of my work. My interest in museums is more personal: I spend considerable time visiting museums and exhibitions, noticing how accessible explanations and engaging experiences can make complex ideas easier to grasp.

Through Yale SOM's Golub Capital Board Fellows Program, I was matched with the Eli Whitney Museum & Workshop. That connection gave me a reason to explore its educational mission.

Founded as a nonprofit in 1976, the museum occupies the historic Eli Whitney Armory site in Hamden, Connecticut. It emphasizes hands-on learning through making, building and experimentation. Its school workshops connect science, engineering and design with practical exploration. That approach resonates with my interest in museums and raised a question: how do teachers connect workshops to classroom goals?

I developed this academic prototype independently, without working with museum staff. The museum has not commissioned, authorized, tested, validated, adopted or endorsed it.

The teacher's decision extends beyond a program name

The museum already offers valuable hands-on workshops and provides a functional website with useful program filters. I explored an additional curriculum-driven planning layer: linking learning objectives to grades, time, program-fee budgets and unresolved questions.

Before Choosing

- What are my learning objectives?
- Which workshop fits my students' grades?
- Does it meet my time and program-fee budget?

Before Visiting

- What still requires museum confirmation?
- How should students prepare?
- How can the experience extend classroom learning?

This was a design hypothesis, not a request from teachers or the museum. The later exploratory proxy-user study offers limited initial feedback; it does not establish educator demand or museum endorsement.

> **The management choice: support judgment, preserve responsibility**
>
> AI can interpret learning goals and suggest activities; deterministic checks enforce verified constraints; teachers select programs and review the plan. My goal is to make planning clearer while keeping uncertainty visible and responsibility with people.

Museum sources: History of the Museum & Workshop; Armory site history; School Programs.

One museum-specific planning workflow.

The contribution is integration: curriculum intent, verified constraints, educational reasoning and an editable plan in one sequence.

Whitney Learning Compass is an evidence-grounded curriculum-to-workshop decision assistant. A teacher describes a goal, compares up to three suitable real programs, makes a choice, and receives Before / During / After teaching suggestions. Review, printing, an editable inquiry draft and browser-local recovery complete the implemented experience.

Reference Product	Documented Capability	Whitney Learning Compass's Distinctive Contribution
Explorable Places	Marketplace for discovering and booking educational experiences across museums and cultural venues, with program details such as grades, subjects, duration, capacity, and pricing.	**Curriculum-driven selection rather than venue discovery.** Starts with a teacher's learning objectives, checks grade, published fees, and duration against explicit constraints, and explains source-backed recommendations from a curated museum catalog. Unverified booking details remain clearly flagged.
MagicSchool AI	AI-powered generation of editable lesson plans using grade levels, instructional objectives, standards, and other teaching requirements.	**Verified workshop selection before lesson generation.** Connects instructional goals to actual museum programs through deterministic eligibility checks. Two tool-using AI agents support program comparison and create an editable Before / During / After learning journey only after the teacher explicitly selects a workshop.
TripSlip	Field-trip coordination platform supporting venue discovery and booking, parent communications, permission slips, payments, and logistics management.	**Educational planning rather than trip administration.** Helps teachers assess curriculum fit, compare verified program constraints and class fees, and develop a workshop-specific instructional plan. It highlights unresolved logistics and prepares an editable museum inquiry without claiming to book or confirm a visit.

> Whitney Learning Compass bridges the gap between discovering a real educational experience and planning how it supports classroom learning. Its contribution lies in combining source-grounded program selection, deterministic constraint validation, explicit teacher choice, and editable instructional planning within one workflow.

These are modest distinctions in scope, based on official product information. They are not a feature-absence audit, paid-product comparison or claim of superiority.

A small catalog, with traceable facts

20
curated real school programs

18
published durations recorded

20
published per-student fees

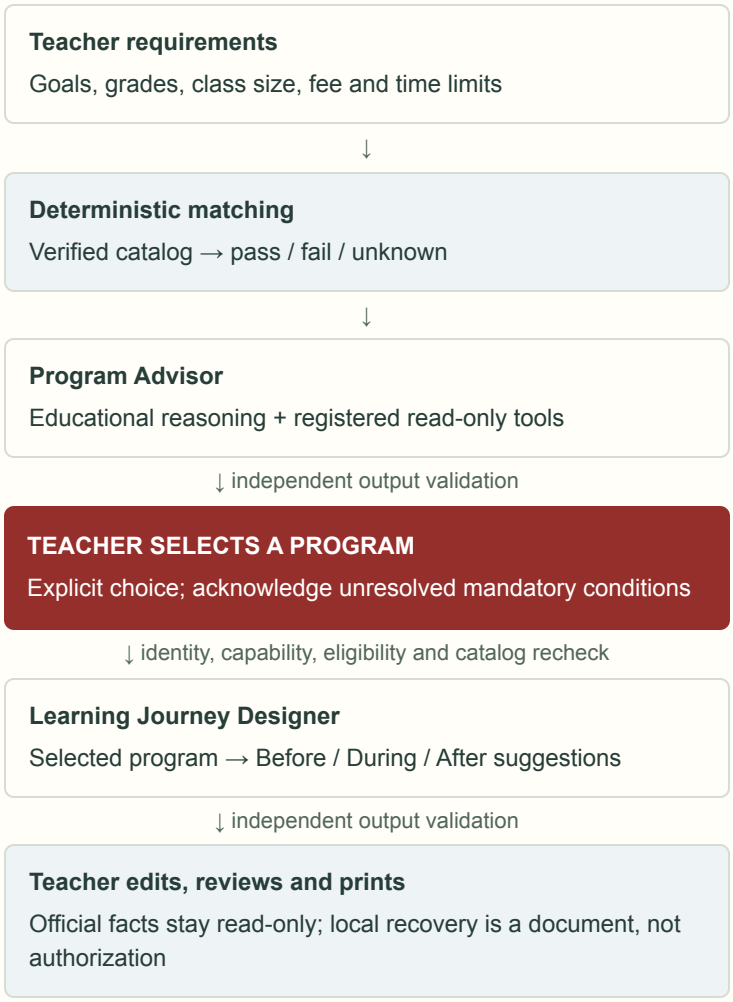
Names, official URLs, original grade labels, subjects and per-student fees are recorded with field-level evidence. USD is a disclosed application assumption for fee calculations, not a museum-published ISO currency code.

Coverage is useful, not comprehensive

The initial audience is Grades 2–7. Grade 2 has only two Engineering and two Sustainability/Energy options in the editorial coverage analysis. Themes can overlap and are not official museum subject labels. The museum's broader catalog remains an important resource.

Sources: Eli Whitney Museum & Workshop School Programs; official Explorable Places, MagicSchool and TripSlip product information. Published facts do not establish current availability, capacity or final trip cost.

Two specialists. One teacher in control.



Program Advisor

Interprets the learning objective and explains educational relevance among real candidates. It can call `search_matching_programs`, `get_program_details` and `get_program_evidence`.

Learning Journey Designer

Receives the teacher-selected program and verified context. Its registered `get_selected_program` tool supplies that program's facts, requirements and unresolved information. It produces structured teaching suggestions for three phases.

The backend owns the sequence

FastAPI orchestrates both PydanticAI agents. Neither delegates to the other. Real catalog IDs, eligibility and authoritative factual fields are checked outside the model; facts are constructed from catalog results rather than trusted model recollection.

An unknown mandatory condition permits only an explicitly acknowledged provisional draft. A failed program cannot become eligible through persuasive prose.

A minimal, demonstrated model connection

Both agents use the course-approved gpt-6-luna model through Portkey's OpenAI-compatible Chat Completions API, with reasoning_effort=none. Real execution traces reported gpt-6-luna-2026-09-22. This configuration was verified through actual tool calls, rather than a basic connectivity check.

The model choice complies with course requirements and fits the bounded, structured task; it is not based on a comparative model benchmark. There is no fallback provider. Credentials remain server-side, with existing environment variables taking precedence and missing values loaded from the project or parent .env files.

> **Two explicit API actions, bounded authorization**
>
> POST /api/recommend returns validated choices; POST /api/journey requires the teacher's selection. Workflows expire after one hour and permit a single journey attempt, even if it fails. A backend restart invalidates them. A restored local draft cannot bypass these checks.

From teaching goal to a validated choice.

A genuine live application run connected the Program Advisor, deterministic eligibility checks, and explicit teacher selection.

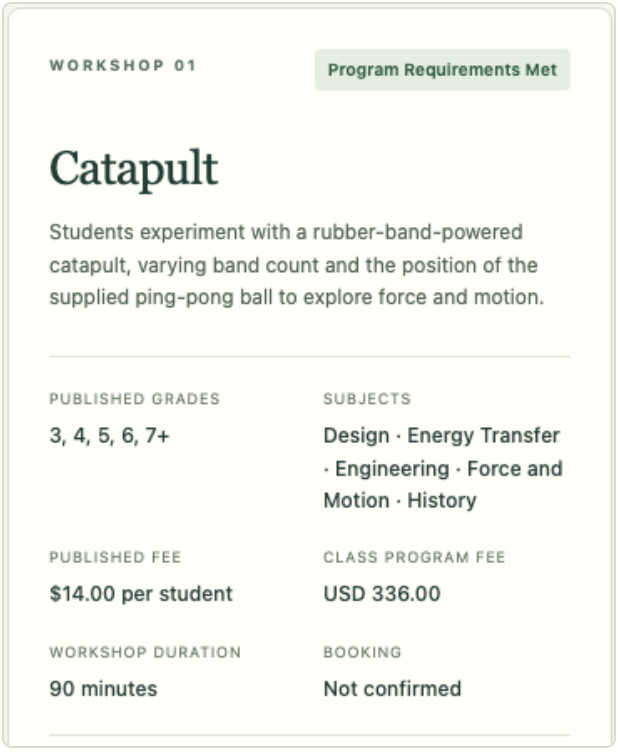
1 · Teacher request

Grade 4 · 24 students · $400 program-fee budget · 90 minutes maximum

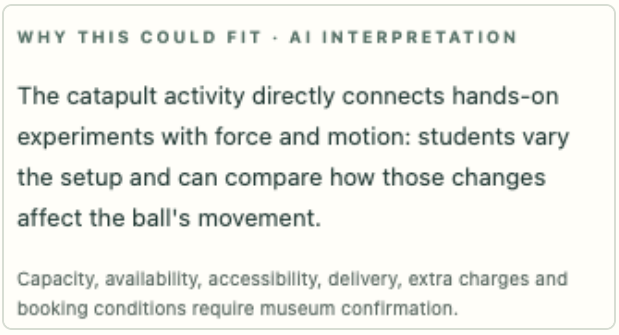
“I want my fourth-grade students to explore force, motion, and engineering design through hands-on experiments.”

Recorded request; USD explicitly acknowledged as an application assumption.

2 · Advisor recommendations



Educational reasoning from the same card



Screenshot from a genuine live AI workflow. Program fees, duration, and eligibility were independently validated. Two unaltered crops of the Catapult card.

Program Advisor — Actual Tool Trace

`search_matching_programs`
↓ 16 verified-catalog candidates
14 pass; 2 have unresolved constraints
↓ `final_result` structured output
↓ 3 independently validated choices

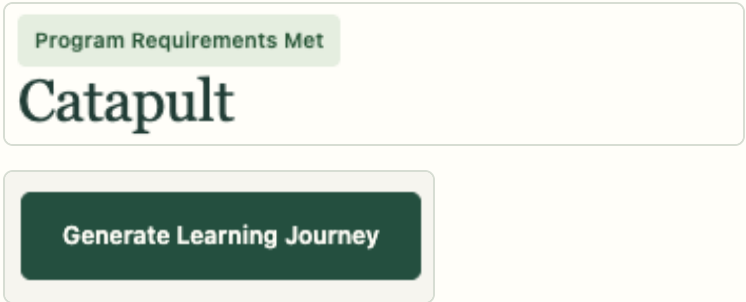
Trace summary derived from the preserved genuine live run.

Recorded recommendations

Catapult · Marble Roller Coaster · The Whitney Relay

Catapult and Marble Roller Coaster each had a $336 class subtotal; The Whitney Relay had $240. All three met Grade 4 and the 90-minute limit. Educational reasoning did not determine those checks.

3 · Teacher selects Catapult



Two crops of the recorded selection review. The teacher’s explicit Catapult choice preceded the separate Designer request.

Eligibility is not a reservation

Availability, capacity, accommodations and extra charges remained unconfirmed. Program-fee subtotals exclude transport, meals and unverified additional charges.

Evidence: successful recorded live workflow, October 22, 2026, after a guard correction; the earlier rejected attempt remains preserved. Official programs: Catapult, Marble Roller Coaster, The Whitney Relay.

From workshop to learning journey.

4 · Learning Journey Designer — Actual Tool Trace

`get_selected_program` → verified Catapult context → `final_result` → validated Before / During / After plan

Trace summary derived from the same preserved genuine live run, after the explicit Catapult selection.

5 · One selected program, three learning phases

BEFORE

Prepare & Wonder

Prepare concepts and predictions; leave observation records blank.

Prepare a simple paper-and-pencil recording sheet with columns for trial or setup, prediction, actual observation, and possible explanation; leave observation fields blank until observations occur.

DURING

Explore & Observe

Follow staff direction and record actual evidence.

Evidence collection

Record the setup and the directly observed result in words or a simple sketch; do not fill in results before observing them.

AFTER

Reflect & Extend

Compare predictions with observations and assess learning.

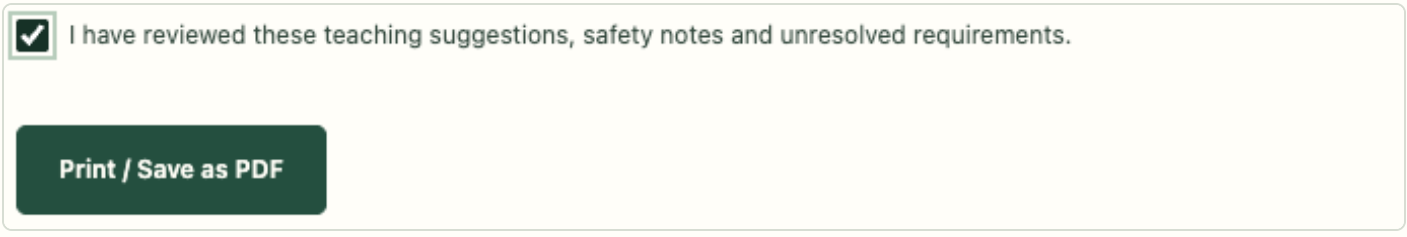
Learning assessment

Objective 1: Review each student's explanation for a connection between a changed setup feature (band count or ball position) and motion, supported by an actual observation. If none is available, assess the student's ability to state what evidence would be needed instead.

Genuine AI-generated learning-journey output for the teacher-selected Catapult program. Suggestions remain editable and require educator review. Three unaltered excerpts from the recorded journey screen.

6 · Teacher Review and Control

Educators can edit the generated learning journey, review its suggestions, and print or save a copy. Browser-local draft recovery preserves editable work without renewing AI authorization or treating saved museum information as current.

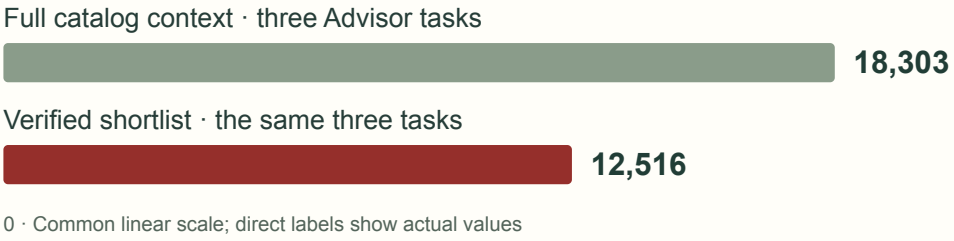


Same live workflow after an objective edit: review acknowledged and a five-page plan printed. The inquiry was edited and copied locally; nothing was sent or booked.

These are teaching suggestions, not official museum curriculum or observed student outcomes. Draft recovery was tested separately offline; no replay image is presented as part of this live run.

Less context helped the Advisor— within a small controlled comparison.

Three scenarios compared **Full** context (20 programs, compact fields) with a **Shortlist** of all nonfailed candidates. Instructions, tools, schema, validators and model settings stayed constant. Full was not the complete archival JSON file.



31.62% fewer Advisor input tokens: (18,303 − 12,516) / 18,303. Six successful Advisor runs; six model requests per condition.

Scenario	Candidates Full → Shortlist	Input Full → Shortlist	Output Full → Shortlist
Grade 4 · force/motion	20 → 16	6,069 → 5,103	347 → 381
Grade 5 · wind	20 → 16	6,074 → 5,076	167 → 169
Grade 2 · geometry	20 → 4	6,160 → 2,337	154 → 156
Total	Three paired tasks	18,303 → 12,516	668 → 706

Recorded Token Usage · Actual Gateway Trace

Grade 4 force/motion · Full catalog · request 2 of 2 · October 23, 2026

input_tokens	cached_input_tokens	cache_write_tokens	output_tokens
5,379	**0**	**5,253**	**332**

Structured excerpt from preserved gateway logs, not a screenshot. Request 1 adds 690 input and 15 output tokens, yielding the table's 6,069 input / 347 output total. Cached reads and writes are subsets of input.

Cost and quality, together

Public-rate Advisor costs: $0.002560700 Full versus $0.001856325 Shortlist, **27.51%** lower. Output rose by 38 tokens. All ten accepted choices had real IDs and correct eligibility; 20 evidence references and 70 authoritative-field checks passed.

Grade 4 choice sets differed. This narrow internal review found no Shortlist prose error; it is not educator validation or proof that filtering preserves quality.

A failure changes the interpretation

Full-context Catapult reasoning misnamed the supplied ping-pong ball as a marble and passed validation. The Full Designer for teacher-selected Marble Roller Coaster was rejected after a guard misread its prohibition on invented observations.

The Shortlist Designer was not run. No paired full-workflow savings or successful new Designer comparison is claimed.

No latency or billing claim. Median Advisor time: 3.088 s Full versus 3.048 s Shortlist; one Shortlist run was slower. Repeated provider response IDs suggest possible response reuse, so neither gateway billing nor fresh-model processing time can be inferred.

These results come from six successful Program Advisor runs in a controlled technical comparison, not from human-user testing. A separate exploratory usability study with one graduate-student participant is discussed in the Impact section.

Small token costs are not a complete operating budget.

One successful historical workflow, measured

Recorded live workflow	Input	Cached read	Output	Requests	Agent time
Advisor	5,127	0	355	2	2.006 s
Designer	4,581	0	887	2	0.552 s
Total	9,708	0	1,242	4	2.558 s

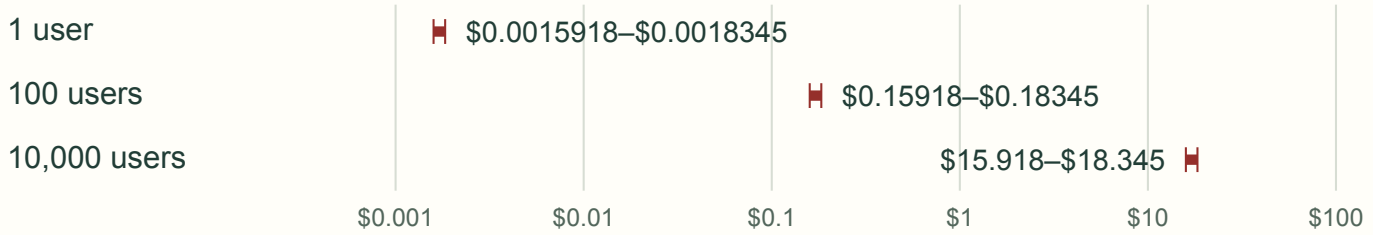
This is summed agent time, not teacher task time. Historical cache writes and response IDs are unknown. Including the earlier rejected pair doubled this transaction usage; that failure remains preserved.

Explicit pricing assumptions

Official Standard USD prices per million tokens: ordinary input $0.10; cached reads $0.01; cache writes $0.125; output $0.50. For total input I, cached reads C, writes W and output O, the illustrative calculation is:

```
[(I − C − W) × 0.10 + C × 0.01 + W × 0.125 + O × 0.50] / 1,000,000
```

Cache classes are disjoint within input. With historical W unknown, W = 0 through I − C gives **$0.0015918–$0.0018345** for this completed example. This is neither a measured course charge nor an estimated average workflow cost.



Logarithmic USD axis · 1 workflow per user/month · public-price projection, not observed billing

Each row repeats one observed example at an assumed usage rate; three workflows per user triples the figures. Hosting, engineering, support, gateway charges and retries are excluded. Actual course billing is unknown.

Bound the spending path before optimizing it

Production limits, per agent: four model requests, six tool calls, 24,000-byte tool results, 2,400 output tokens per response; 90 s agent and 30 s HTTP timeouts; zero SDK/output retries. A two-agent workflow can require up to eight requests.

Additional guards: 70,000-byte HTTP requests and cumulative reported usage thresholds of 32,000 input / 6,500 output tokens; a completed response can overshoot a cumulative threshold. Editing, comparison, printing and autosave add no inference.

The controlled evaluation used **14 of 24 authorized model requests**: 35,388 input, 0 cached reads, 29,280 writes and 2,295 output tokens; illustrative cost $0.0054183. Eleven distinct response IDs suggest reuse, without establishing billing.

Pricing sources: OpenAI's official GPT-6 Luna pricing and prompt-caching documentation. All costs shown are illustrative estimates based on public token rates, not actual course-gateway charges.

Recommendations must pass the rules.

Defense Test · Budget Override Blocked

Grade 4 · 24 students · 90-minute limit · USD assumption disclosed.
Catapult: 24 × $14 = $336.00.

Teacher budget	Verified program-fee subtotal	Simulated model proposal
$335	**$336.00**	**eligible**
Independent matching	**Validator result**	**Recommendation accepted**
fail	`invalid_model_output`	**No · rejected before return**

Faithful rendering of the executed test receipt. Recorded rejection reason: “Eligibility override”. Test outcome: passed.

This was an offline adversarial test using a simulated model response and the application's real validation logic. It was not a live-model attack.

This test demonstrated that a program cannot satisfy a verified constraint merely because the model considers it educationally relevant.

Decision	Rule	Concrete implication
Fail	A verified hard requirement is violated.	Grade mismatch remains fail despite unknown duration.
Unknown	No verified failure; a mandatory fact is unverified.	Bridges or Rubber Band Car needs confirmation under a duration limit.
Pass	Applicable requirements are satisfied; assumptions disclosed.	Missing duration alone does not block a request with no time limit.

Protect the whole decision path

Risk	Implemented boundary / evidence
User or source instruction injection	Tested override/secret requests rejected; malicious source mutation blocked before inference. Catalog data cannot authorize instructions.
Fabricated or cross-workflow selection	IDs, membership, capability, expiry and catalog hash rechecked. Expired or mismatched selections blocked.
Unconfirmed operations or invented costs	Capacity, booking, accessibility and additional charges stay unconfirmed. USD is assumed; transport and meals are excluded.
Invented empirical evidence	Label predictions; leave observations blank until collected. Suggestions do not imply an experiment happened.
Sensitive browser persistence	Allowlisted drafts exclude credentials, capabilities and private traces. Restore is document editing, not authorization.

The controls narrow risk; they do not establish universal injection resistance or factual accuracy. Natural-language claims still need scrutiny, as the actual failures on the next page show.

The demonstrated defenses are offline technical tests of actual validation logic, not user studies. Program source: official Catapult page.

Two observed errors led to two bounded repairs.

The most informative findings were two validation failures: one incorrect claim was accepted, while one appropriate warning was rejected.

Observed Validation Failure	Targeted Validator Fix	Verified Offline Result
False negative: Full Advisor called Catapult's projectile a marble; the output was accepted.	Program-specific material guard checks Catapult reasoning against its documented supplied ping-pong ball. A different program named Marble Roller Coaster remains valid.	The exact preserved Advisor output is now rejected.
False positive: Full Designer advised against filling gaps with invented measurements; the guard treated this prohibition as fabrication.	Clause-aware checks evaluate each fabrication expression with its local prohibition context; a global "not" cannot excuse unrelated fabrication.	The exact preserved Designer output is now accepted without rewriting its text.

The original traces remain unchanged. Offline acceptance of the old Designer text is **not** a new successful live Designer run and does not complete the missing paired experiment. The repairs improve specific known cases without pretending to solve semantic factual grounding.

What remains fragile

Catapult's guard is program-specific. The empirical-evidence guard still relies on bounded language patterns. Synonyms, multilingual text and complex negation can produce missed or overblocked claims.

Future evaluation should broaden these cases while preserving the nonnegotiable prohibition on fabricated observations. A generic factual-entailment guarantee is not established.

What was actually validated

Previously executed tests included **284 passing Python tests**, **153 passing frontend tests**, and six passing evaluation-harness tests, along with successful TypeScript checks and a production build.

These results come from previously executed regression and evaluation tests. They provide evidence for the specific cases tested, not a guarantee of complete factual accuracy or safety.

Frozen offline evaluation	Result	What the denominator means
Predefined technical scenarios	24 / 24	Ordinary, conflicting, incomplete and adversarial requests.
Specified eligibility decisions	33 / 33	Enumerated program decisions; includes multiple decisions in one scenario.
Hard-constraint assertions	11 / 11	Specified assertions, not eleven independent people.

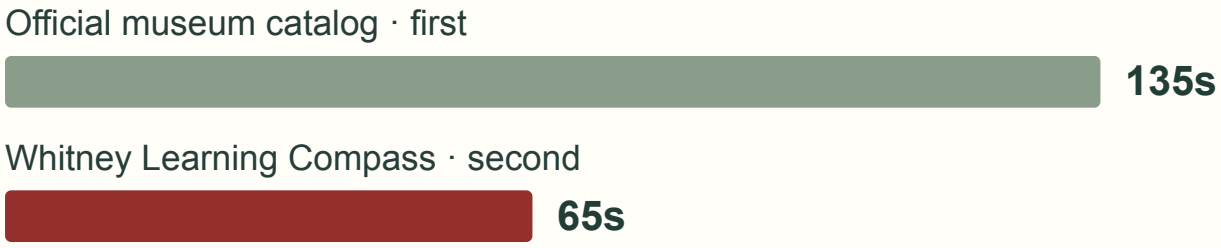
The first harness execution passed 23 of 24 scenarios because a UUID/string lookup failed in the test harness. That harness issue was repaired and the full suite passed; the original failure was retained. These controlled results do not measure classroom usefulness or establish a population-wide accuracy rate.

Evidence: Original model outputs and before-and-after offline validation tests were preserved. Both fixes were verified without additional model calls.

Exploring a More Efficient Path to Museum Program Selection.

On October 23, 2026, I asked a Yale SOM graduate student to test Whitney Learning Compass against the Eli Whitney Museum & Workshop's official school-program catalog. The participant had no K–12 teaching experience and served as a proxy user. They completed the same Grade 4 planning scenario, using the official catalog first and Compass second.

The scenario specified 24 students, a maximum $400 program-fee budget under the disclosed USD assumption, a maximum 90-minute workshop, and a learning objective of force, motion and engineering design.



70 seconds less active selection time. (135 − 65) / 135 = 51.85%, rounded to 51.9%.

> **51.9% less measured active selection time in one exploratory proxy-user task.**
>
> Timing excluded AI waiting, learning-plan review and questionnaire completion. This is not total planning-time savings, an average teacher benefit or a causal estimate.

All five participant ratings

All ratings use a five-point scale, where 1 is the lowest rating and 5 is the highest.



Single-participant subjective ratings on a five-point scale. No composite satisfaction score or statistical inference is calculated.

Both interfaces produced a suitable choice	Official catalog	Compass
Selected program	Simple Machines	Marble Roller Coaster
Published fee / class subtotal	$14 / $336	$14 / $336
Published duration	90 minutes	90 minutes

Neither choice confirmed booking/capacity. Fixed order may favor Compass; comparative instructional quality was not independently evaluated by an educator.

Convenient comparison. A useful draft that still felt generic.

The feedback supports further testing of integrated discovery. It also shows that free-text input and AI-generated plans need additional refinement.

What worked for the participant

Information in one place

Fees, duration and unresolved operational questions were useful together. The participant perceived less need to assemble information across pages; page visits were not separately counted.

A starting point for an inquiry

The editable museum inquiry draft was convenient. No separate email-writing time was measured, and the app did not send the inquiry.

Compass discovery/comparison and confidence both received 4/5. Willingness to use for planning was also 4/5. These are perceptions from a proxy user, not adoption or educator-performance measures.

What needs improvement

More instructional specificity

The learning journey was useful as an initial outline but felt generic, reflected in its 3/5 usefulness rating. Detailed activity design and teacher judgment remain necessary.

More guidance at intake

The museum's predefined browsing filters made categories easier to understand. Compass primarily used an open-ended learning-goal field, leaving the participant unsure which topics or wording to start with. They suggested selectable subjects or keywords alongside optional free text.

Existing Quick Start examples are implemented. They do not constitute the broader guided curriculum-input interface suggested by the participant.

How much weight should this evidence carry?

This is one exploratory comparison with a graduate-student proxy, not a K–12 teacher. The fixed website-first order cannot isolate interface effects from learning or familiarity. Timing excludes AI waiting and later plan review. No classroom delivery, student learning, booking or museum adoption was measured. These findings do not establish a generalizable productivity benefit.

These observations and ratings were documented from a real usability session with one Yale SOM graduate-student proxy participant. The findings reflect the participant's recorded feedback and are presented as paraphrases rather than direct quotations.

The pilot suggests value in connecting information; it does not remove the need for educator judgment.

Evidence: Exploratory usability test with one Yale SOM graduate-student participant, October 23, 2026.

The next step is better validation with the people who would use it.

A fellowship connection and curiosity about hands-on learning led to a working independent academic prototype. The pilot participant found value in integrated program comparison, while feedback on the generic learning plan and open-ended input highlighted opportunities for improvement.

If another month were available

1 · Test with real educators. Co-design with teachers and, with permission, museum stakeholders. Counterbalance order and measure end-to-end time, correction effort and instructional usefulness.

2 · Guide curriculum discovery. Offer selectable subjects, curriculum themes and topic suggestions alongside optional free-text goals. This would address the participant's difficulty starting with an open-ended field.

3 · Make learning journeys more customizable. Explore selectable levels of instructional detail and adaptable Before / During / After activities. Develop richer prompts with educator review; keep suggestions editable and distinct from official museum curriculum.

4 · Broaden evidence and strengthen validation. Expand verified workshop coverage and establish source-refresh processes. Investigate standards mappings only with authoritative support, flag incomplete or outdated information, and broaden factual, safety and negation tests. Never infer missing facts.

5 · Support follow-through and assess feasibility. Explore a checklist covering availability, capacity, accessibility, materials and transportation, without confirming reservations. Before deployment, evaluate operating needs, verify gateway billing, complete a paired Designer study and seek organizational review.

Limits of the current product

A curated 20-program catalog has incomplete grade/theme coverage. Source review does not establish current availability, capacity, accommodations or final pricing. Two durations and all program-specific standards remain unpublished in the dataset.

Natural-language validation remains heuristic. Local drafts are browser-specific historical documents. The prototype is not deployed or adopted by the museum.

Limits of the current evidence

One non-teacher proxy, fixed order and excluded AI waiting prevent a general productivity claim. Generic instructional plans require editing. The paired experiment supports Advisor-context savings only; course billing and full-workflow optimization remain unresolved.

No museum endorsement, classroom impact, nonprofit reach or student-learning gain has been measured.

What I learned: a credible AI planning assistant is more than a good answer. It is a sequence that makes evidence inspectable, uncertainty actionable, and the educator's final decision explicit.

Future directions are proposed extensions, not implemented features or museum-approved plans.